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Decentralized Energy Systems

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EnergisedFutures
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Agenda

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- Energised Futures

Local Energy Systems

- Energy Communities
- Local Energy Markets

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Introducing Centrica

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Centrica

Centrica is a leading international energy services and solutions provider, founded on a 200-year heritage of serving people. We serve around 10 million customers across the UK, Ireland and continental Europe through brands such as British Gas. Centrica plc is a constituent of the FTSE 100 on the London stock exchange (CNA) and holds a BBB credit rating.

21,000	Colleagues in our team
10+ m	Total customers
16.7 GW	Renewable capacity under contract
> 50 %	Of the UK's total gas storage capacity
20 %	Share of the UK's nuclear portfolio

Centrica Ecosystem

British Gas

centrica
Business Solutions

centrica
Energy

Bord Gáis
Energy

centrica
energy storage+

SPIRIT
ENERGY

Centrica's Net Zero Commitment

We're helping to create a more **sustainable** and **inclusive** future that supports **our communities, our planet** and **each other**

We have established People & Planet targets…

Be a net zero **business** by 2045

40%

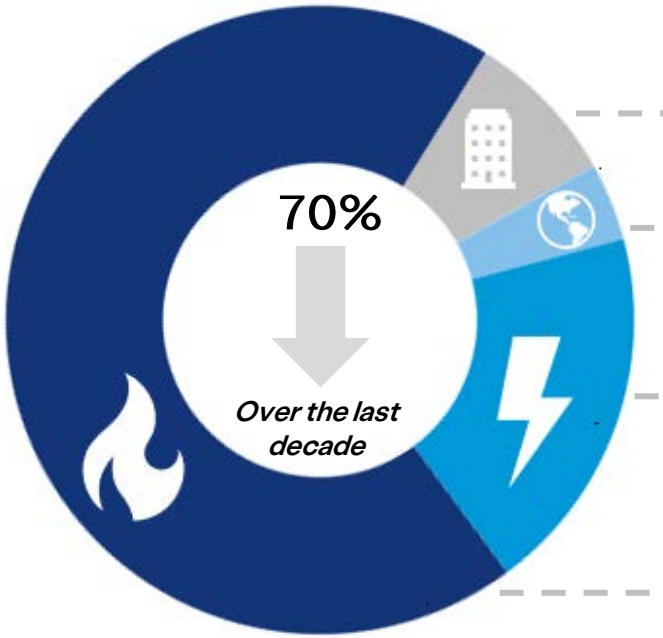
carbon reduction by 2034

Help our **customers** be net zero by 2050

28%

carbon reduction by 2030

…to reduce Centrica's global GHG footprint (Scope 1,2 & 3)



Centrica target

Centrica 2mt CO_2e	8%
Supply Chain 1mt CO_2e	4%
Elec sales 5mt CO_2e	19%
Gas sales 18mt CO_2e	69%

How does Energised Futures fit into all this?

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Who we are

An in-house research and innovation incubator launched to accelerate a new energy future and catalyse Centrica's purpose to energise a greener, fairer future.

Our team

In 2025

In 2026

We have 6 pillars (today...)

Flexibility

how smart tech and consumer behaviours can change supply and demand

Automation

how to optimise device control at scale

Markets

how to better value, settle, balance, and regulate energy

Connectivity

how to reliably connect properties and new devices

Engagement

how to positively impact perceptions and adoption of new energy solutions

Accessibility

how to ensure the future of greener and fairer energy is inclusive of all

We're delivering a range of projects

How will bidirectional EV charging change consumption and grid pressures?

Can sharing energy between neighbours reduce bills and increase access?

With limited data and customer constraints how do we model and deploy flexibility?

Can we adapt to increased data availability and security risks and create a next-gen platform?

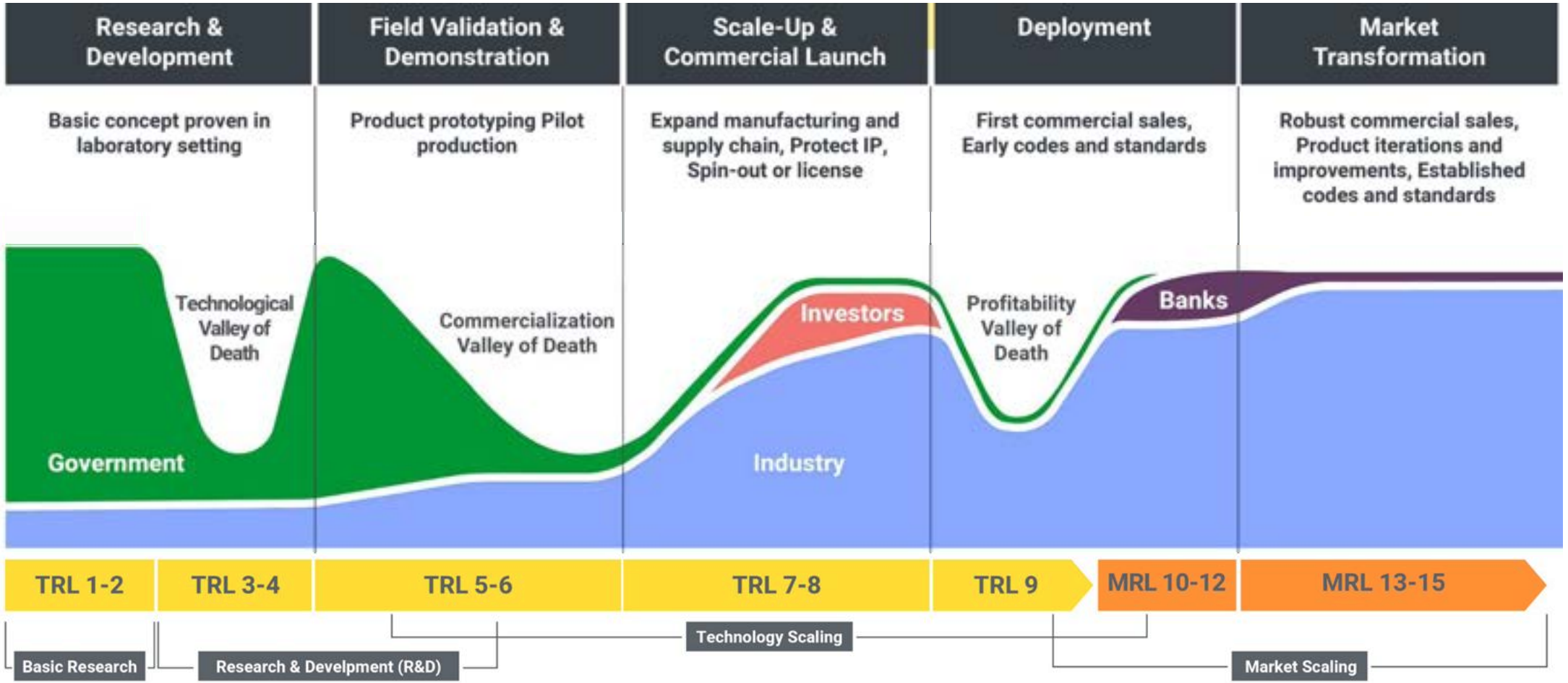
Resilient

Reliable

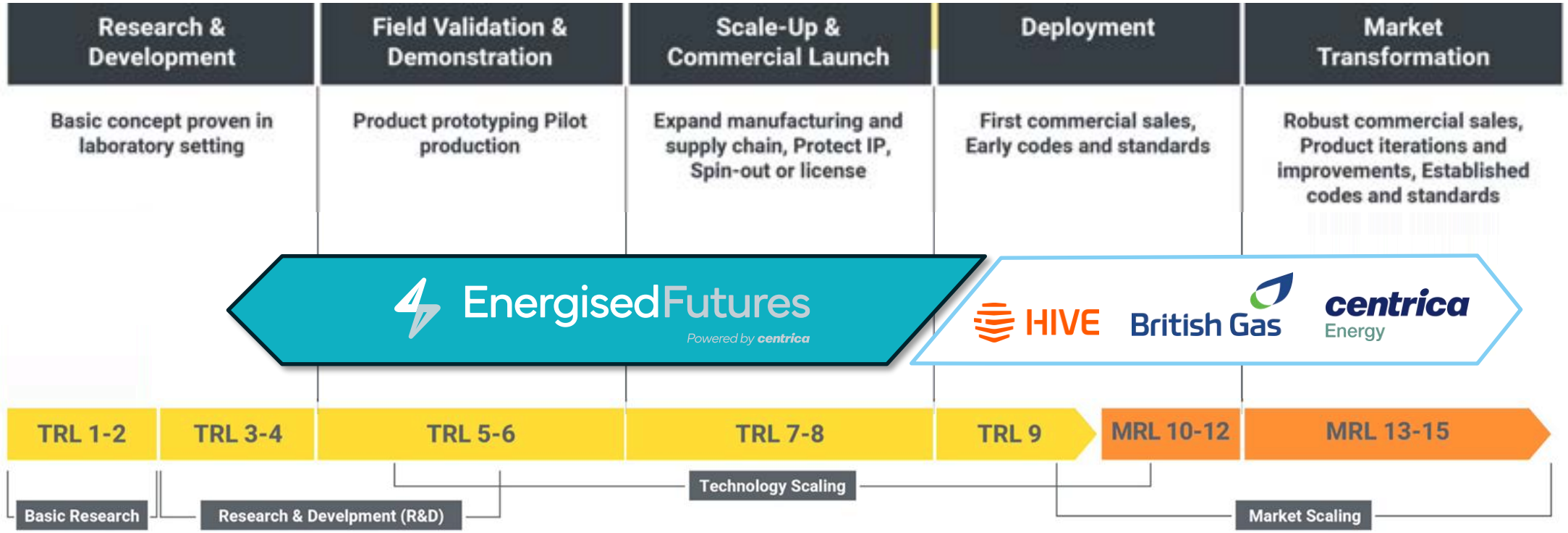
Diversified

Democratised

Technology and market readiness levels



Research-led growth

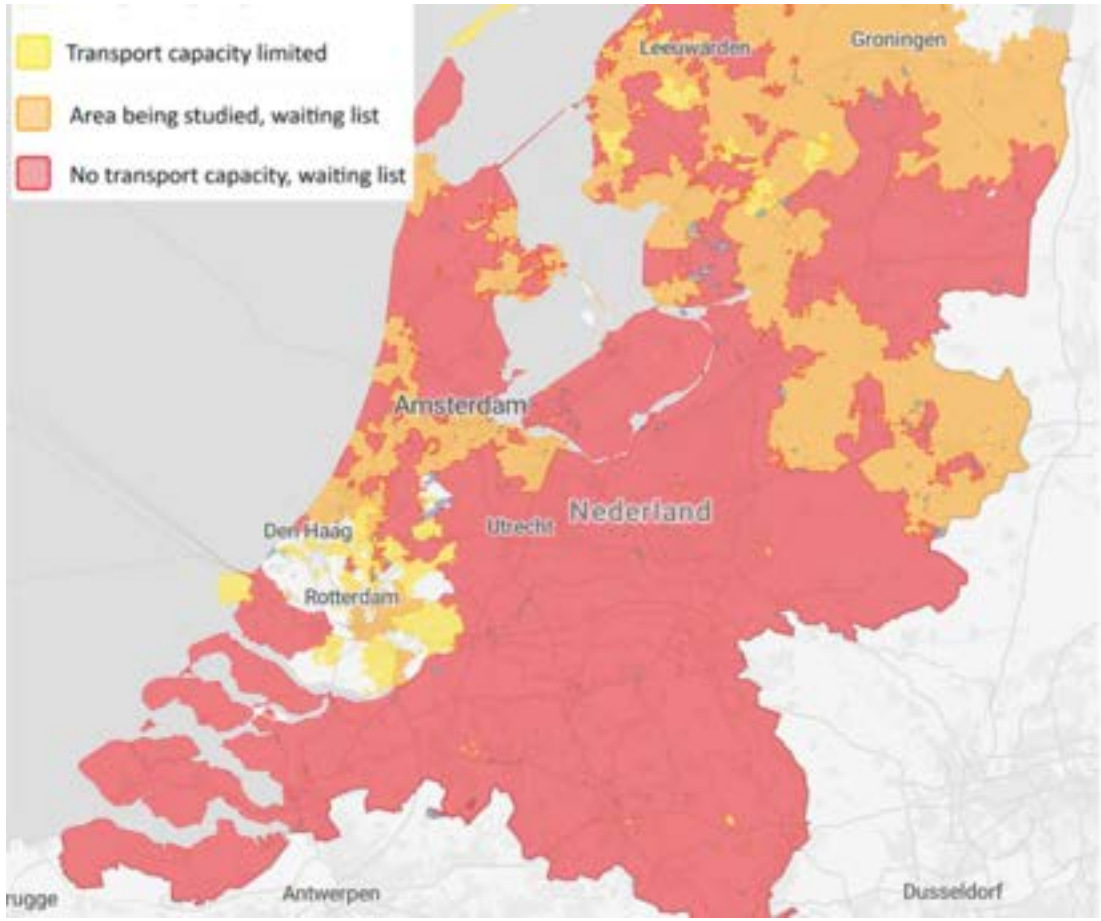


Disruptive innovation starts with big ideas and deep understanding of problem and context. But bringing an idea from the back of an envelope to a mature mass market take a long chain of funders and a broad ecosystem of innovation supporters. We're Centrica's route through and root into in this ecosystem.

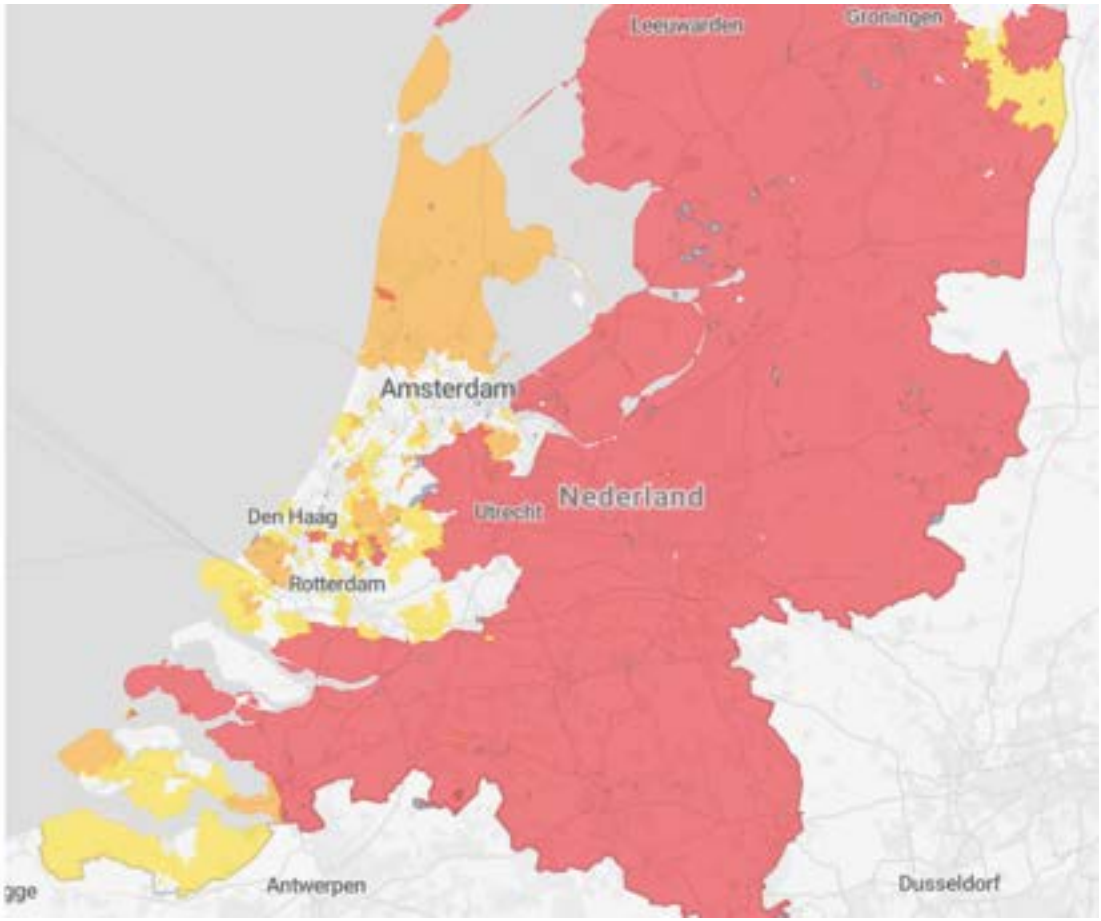
Decentralising Energy Systems

Grid congestion is a very real threat

Grid congestion in the Netherlands (December 2024)

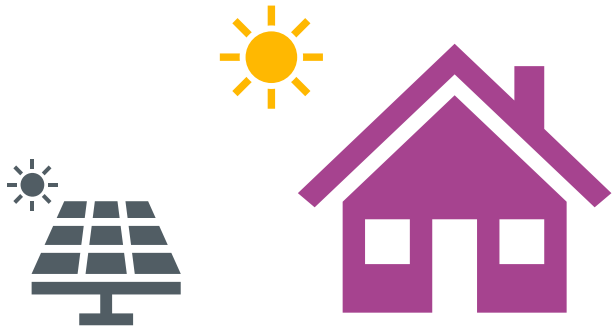


Grid congestion for electricity demand



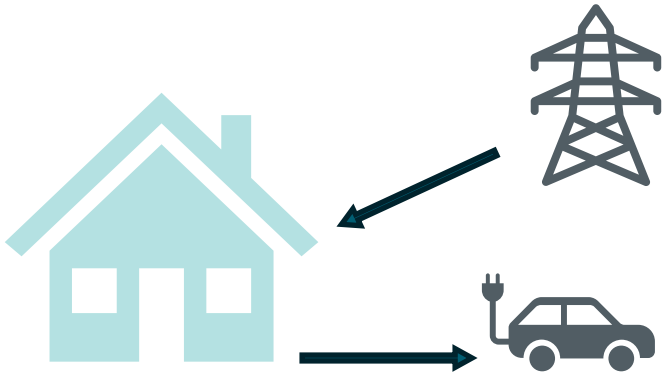
Grid congestion for electricity supply

An example: smart power today



Home A

- Generating electricity from PV
- Uses this to charge **battery**
- Any further excess exported to grid, receives **feed-in tariffs**



Home B

- Not generating enough to meet **demand**, or has **no generation assets**
- Waits to charge EV overnight, **importing** electricity from grid
- Pays **unit rate** for electricity imported

Challenges of current energy systems

Grid Capacity

- The grid was designed for unidirectional flow (generator to consumer)
- Additional generation and load create congestions
- All of this may lead to renewables being curtailed

Costs

- There is usually a large gap between high retail prices and low feed-in tariffs
- Loss of revenue from renewable curtailment

Barriers

- High capex necessary for a fully-electric, smart home is a barrier for many to access clean energy

How could we be using energy ?

- Generating **excess** electricity from PV
- **Sends** excess electricity to **Home B** rather than charging battery
- Receives **higher** smart export guarantees

- Not generating enough to meet **demand**, or has no **assets** to generate
- **Receives** electricity from **Home A**, which is used to charge EV immediately
- Pays **lower rate** for electricity received

Local Energy Initiatives

Renewable Energy Communities (RECs)

A group of citizens, usually located in the same geographical area, investing in co-owned assets, or sharing energy generated by their own assets

Local Energy Markets (LEMS)

A decentralised platform where local energy producers, consumers, and prosumers trade energy and flexibility services.

Increasing access to cheaper, greener energy

Benefits of Renewable Energy Communities include:

Pooling assets extracts more value whilst reducing upfront costs

Providing access to renewable generation to a larger number of people through energy sharing

More people adopt smart / eco-tech solutions, hitting government Net-Zero targets for new builds

	Individualistic approach	Energy Community
Upfront cost	High	Lower
Savings	High	Higher
Energy cost	Low	Lower
Effort	High	Low

Can RECs member earn more money?

Yes, LEM!

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Distribution Service Operators (DSOs) are experiencing new grid needs.

Flexibility-based services have been defined to solve those

Traditional Solution:

- Grid Reinforcement
- Tap changers
- Grid expansion
- Curtailment

New solution:

✓ activation of the flexibility (both upward and downward)

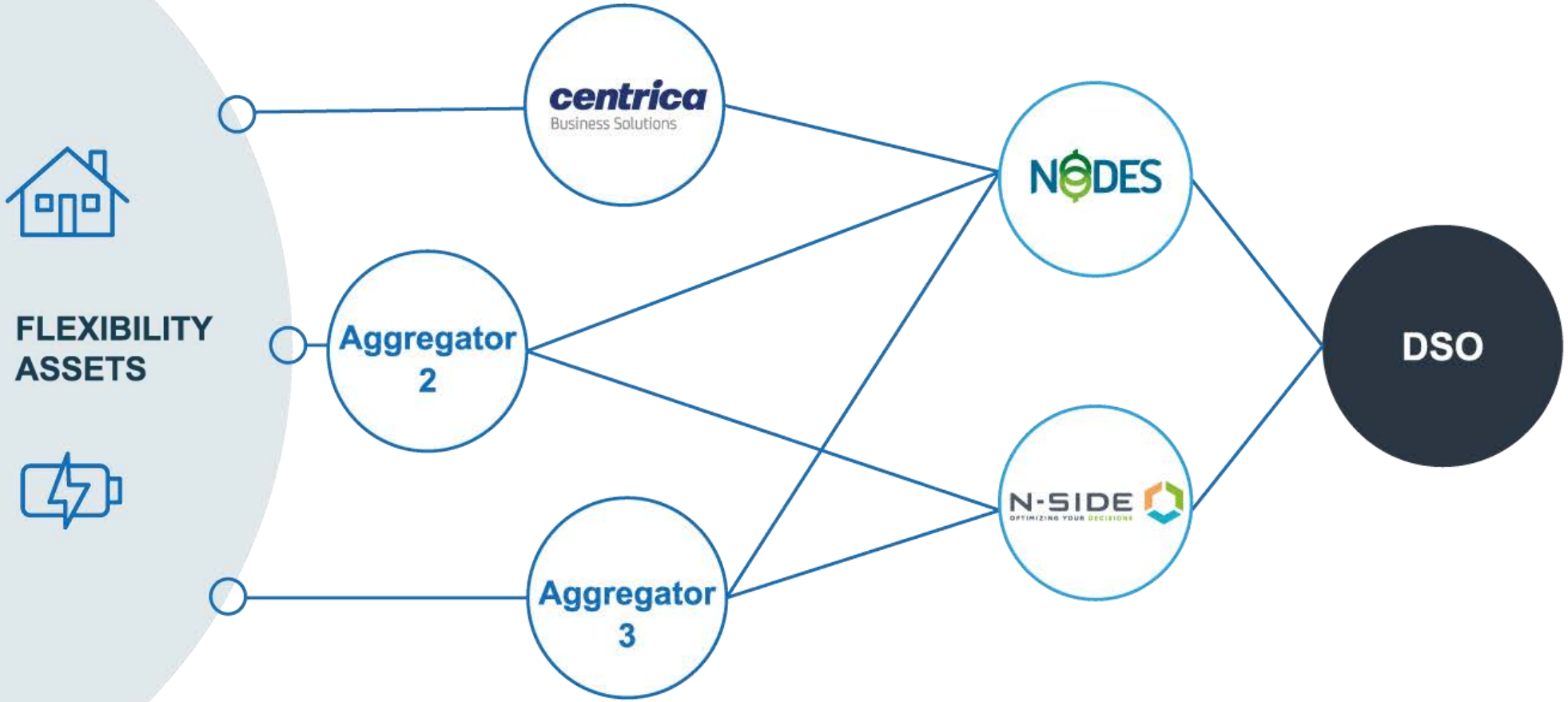
Flexibility is its own market

Product: Flexibility (power, reactive power, energy)
Service: Provision of flexibility for grid services

NODES GOPACS epexspot N-SIDE piclo

Flexibility: Energy system flexibility is the ability to adjust supply and demand to achieve/maintain the balance of the energy systems

Local Energy Market (LEM) players



Our Journey in LEM/LFM: Past, Present & Future

Cornwall

The Centrica LEM project enables coordinated flexibility trading between local and national grid operators via a single platform, helping balance the grid and integrate renewables by rewarding users for adjusting energy use

EUniversal

EUniversal develops a standardised interface (UMEI) to help DSOs use energy flexibility and connect with new markets across Europe

LocalRES

LocalRES aims to demonstrate the ability of RECs to provide energy services and maximise flexibility

Neighbourhood Flex

Neighbourhood Flex seeks to take the learnings and technology from LocalRES and apply them to a UK context

What matters in LEM

Consumer engagement

Needs and products

External coordination

Standardised communication set-up

Data transfer in function of stakeholders' responsibilities

Planning (Operation & network investment)

AI in energy systems

Forecasting application

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Applications

Business

Back-testing

Test how well control strategy performs in the past

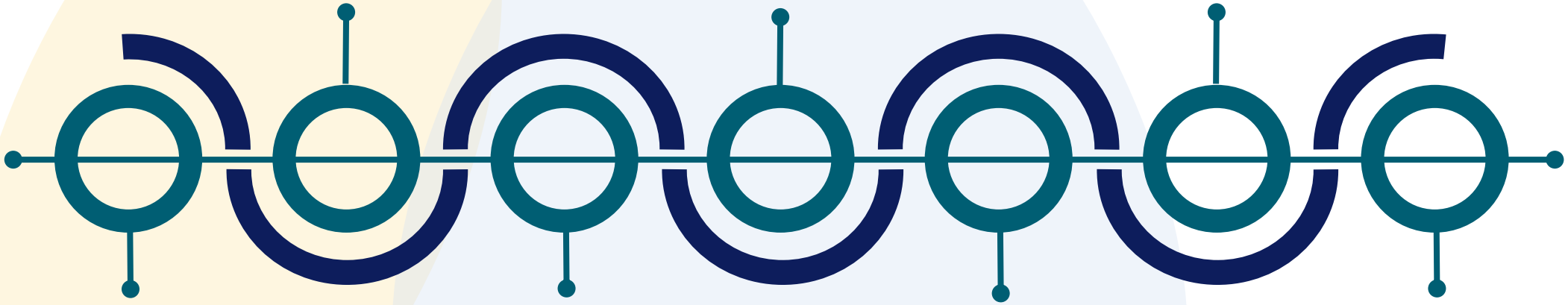
Optimisation & Trading

Forecasting

Predict the behaviour of the assets given historical data

Simulation

Verify the integrity of the IoT solution on a simulated asset



Asset model

Construct a digital version of the assets to be tested and simulate with perfect knowledge

System Identification

Identify the behaviour of the system based on the measurements

Asset Optimisation

Extract optimal control strategy based on asset constraints

Online Control

Monitor the assets in real-time and update their controls

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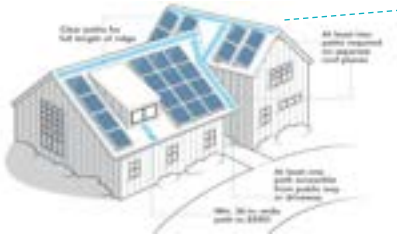
Use Case 1: Solar Forecasting for RECs

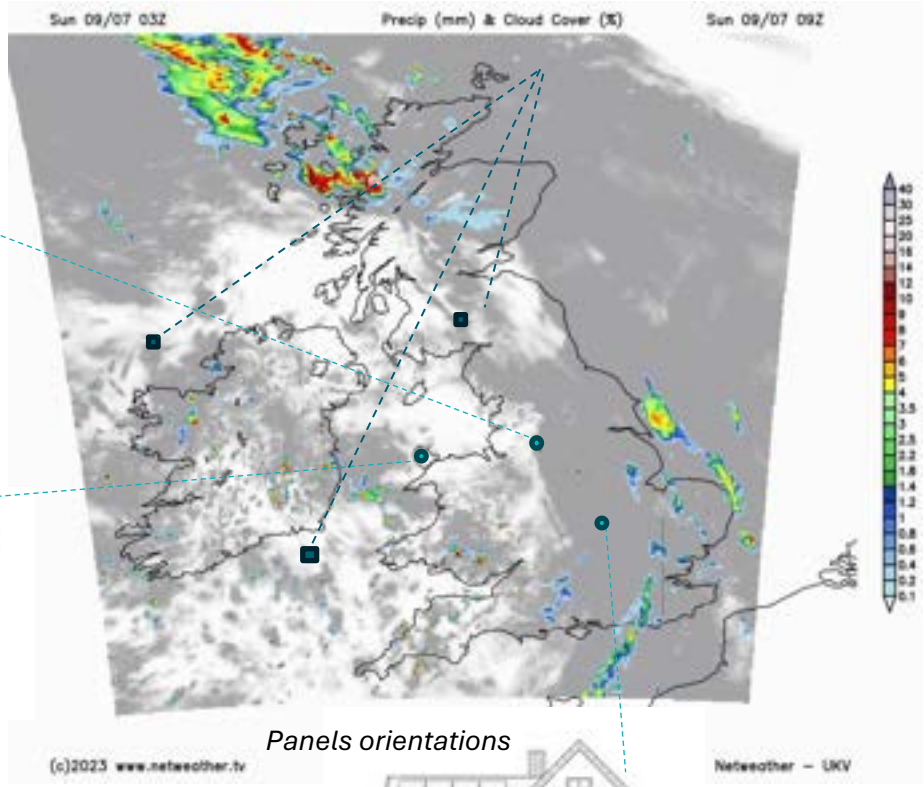
How can we leverage the solar PV data of thousands of existing customers to accurately predict solar production for new customers?

Weather conditions

Shading

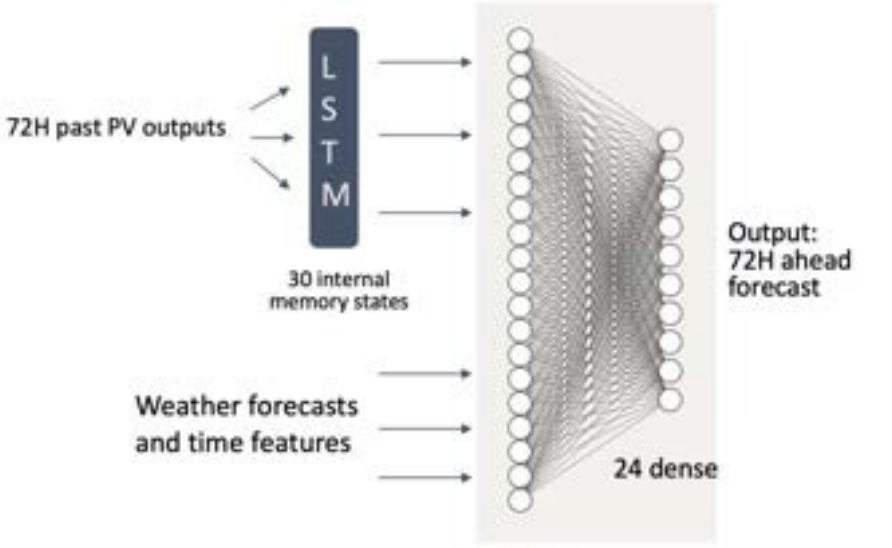
Array composition

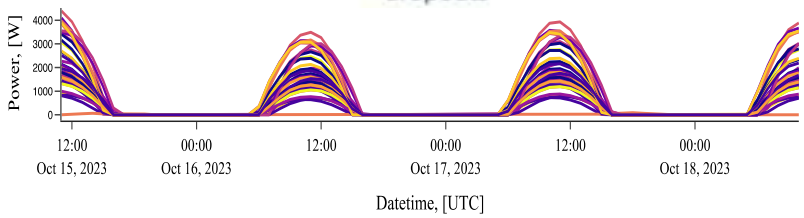




Panels orientations

Forecasting model





Use Case 2: Uncertainty Forecast

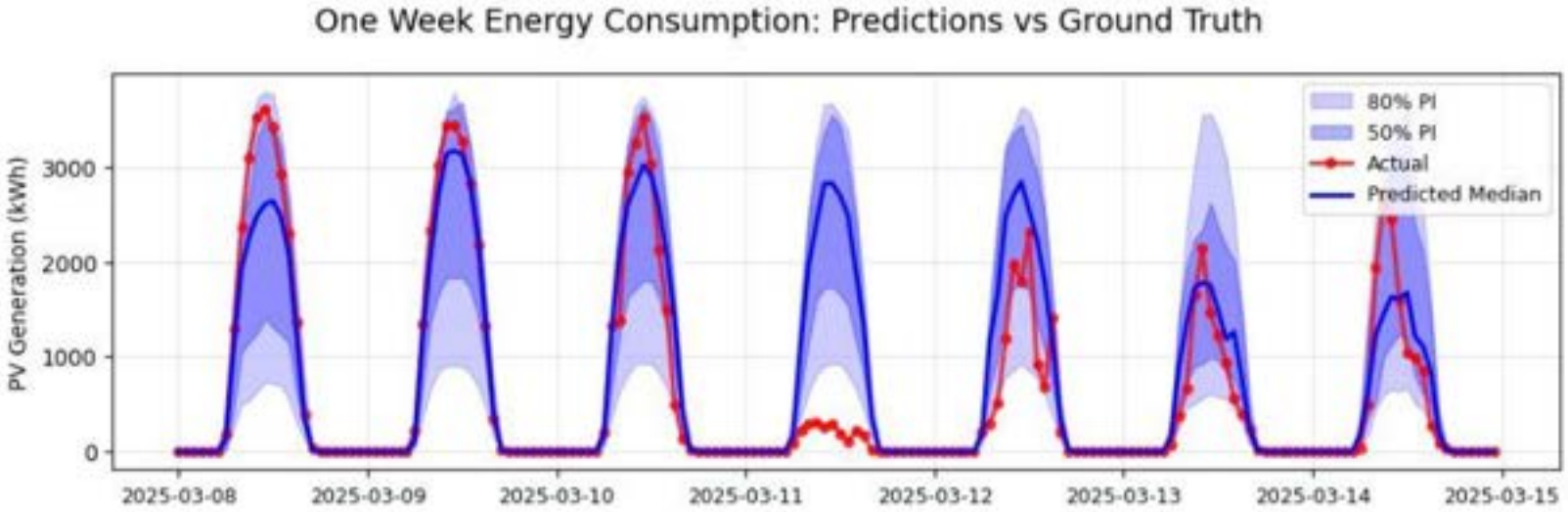
Dataset Description

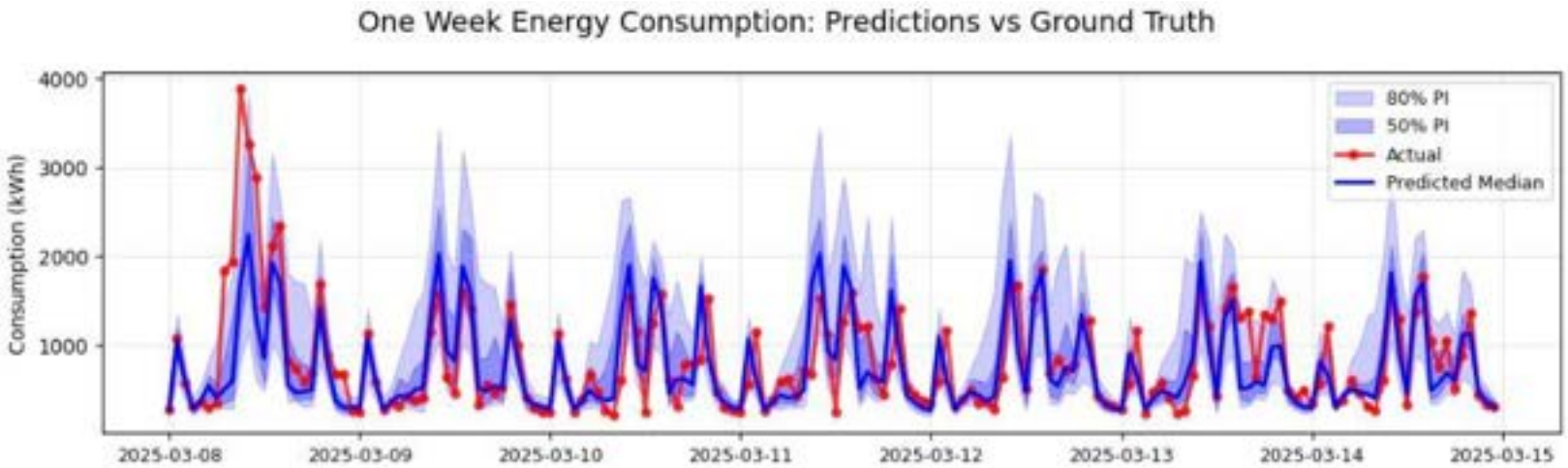
- Residential energy measurements capturing demand and PV generation
- 295 households
- Extends from January 2024 through April 2025 for a total of 488 days
- Split as 60% for training, 20% for validation, 20% for testing
- Forecast horizon is 48 hours

Methodology

- Probabilistic forecasting using quantile regression
- The model predicts nine quantiles (1%, 5%, 10%, 25%, 50%, 75%, 90%, 95%, 99%) for each forecast horizon
- Employs LightGBM with quantile regression as the core predictive algorithm
- Features:
  - shifts of 48, 72, 96, 120, 144, and 168 hours,
  - hour of day and day of week
  - temperature is used for consumption forecasting due to its influence on heating demand

Use Case 2: Uncertainty Forecast





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